Model Selection for Reliability Estimation in Series Systems

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Abstract

This paper explores model selection for reliability estimation of components in multi-component series systems using Weibull distributions. We assess the sensitivity of a likelihood model based on right-censored and masked failure data to deviations from well-designed series systems where components have similar failure characteristics. We compare a full model with heterogeneous component shape parameters against a reduced model assuming homogeneous shapes, which renders the system Weibull and reduces parameters from $2m$ to $m + 1$. Appropriateness of the reduced model is assessed using likelihood ratio tests across varying sample sizes and component configurations. For well-designed systems, the reduced model maintains excellent fit even with sample sizes exceeding 10,000 observations, requiring approximately 30,000 observations before rejection in 95% of simulations. Deviations in individual component scale or shape parameters quickly provide evidence against the reduced model. Maximum likelihood estimators demonstrate sensitivity to system design but remain robust under moderate masking and censoring. Proper model specification requires balancing simplicity against representativeness based on system characteristics and available data.

Contents

1 Introduction

Estimating reliability of individual components in multi-component systems is challenging when only system-level failure data is observable. A likelihood model incorporating right-censoring and candidate sets was previously developed to enable component inference from such masked data [1]. Simulation studies revealed estimator behavior and sensitivity to modeling assumptions given limited samples.

A key question is choosing an appropriate model complexity. A reduced model assuming homogeneous component shapes simplifies analysis as the system becomes Weibull. However, deviations in component properties impact adequacy. Proper model specification requires balancing simplicity against representativeness.

This paper explores model selection for component reliability estimation in series systems. The likelihood model from [1] is summarized. Simulation studies demonstrate estimator sensitivity and assess reduced model appropriateness using likelihood ratio tests. Findings provide guidance on suitable models based on system design and available data.

The remainder of this paper is organized as follows. Section 2 presents mathematical preliminaries including formal notation, series system definitions, Weibull distribution properties, and foundational theorems. Section 3 summarizes the likelihood model for masked and censored data. Section 4 presents simulation studies assessing estimator sensitivity to system design variations. Section 5 introduces the reduced homogeneous shape model and evaluates its appropriateness using likelihood ratio tests. Section 6 concludes with practical guidance on model selection.

2 Mathematical Preliminaries

2.1 Notation and System Structure

Consider a series system composed of m components, where the system fails when any single component fails. We observe n independent and identically distributed system lifetimes. For the i-th system ($i = 1, \ldots, n$), let T_{ij} denote the lifetime of component j ($j = 1, \ldots, m$). The system lifetime is given by

$$T_i = \min\{T_{i1}, T_{i2}, \ldots, T_{im}\}. \tag{1}$$

We denote the complete parameter vector by $\boldsymbol{\theta} = (k_1, \lambda_1, k_2, \lambda_2, \ldots, k_m, \lambda_m)$, where $k_j > 0$ is the shape parameter and $\lambda_j > 0$ is the scale parameter for component j. Bold symbols represent vectors throughout this paper.

2.2 Weibull Distribution

Each component lifetime follows a two-parameter Weibull distribution with probability density function

$$f_j(t; \lambda_j, k_j) = \frac{k_j}{\lambda_j}\left(\frac{t}{\lambda_j}\right)^{k_j-1} \exp\left\{-\left(\frac{t}{\lambda_j}\right)^{k_j}\right\}, \quad t > 0, \tag{2}$$

reliability function

$$R_j(t; \lambda_j, k_j) = \exp\left\{-\left(\frac{t}{\lambda_j}\right)^{k_j}\right\}, \tag{3}$$

and hazard function

$$h_j(t; \lambda_j, k_j) = \frac{k_j}{\lambda_j}\left(\frac{t}{\lambda_j}\right)^{k_j-1}. \tag{4}$$

The mean time to failure (MTTF) for a Weibull-distributed component is

$$\text{MTTF}_j = \lambda_j \Gamma\left(1 + \frac{1}{k_j}\right), \tag{5}$$

where $\Gamma(\cdot)$ is the gamma function. The shape parameter k_j characterizes the failure mode:

- If $k_j < 1$, the hazard function decreases with time, indicating infant mortality or early-life failures.
- If $k_j = 1$, the hazard function is constant, corresponding to an exponential distribution with memoryless failures.
- If $k_j > 1$, the hazard function increases with time, indicating wear-out or aging failures.

2.3 Data Structure

The observed data for each system i consists of:

- **Observed system lifetime** t_i: The time at which system i fails or is censored.
- **Censoring indicator** δ_i: Equals 1 if system i failed and 0 if right-censored.
- **Candidate set** $C_i \subseteq \{1, 2, \ldots, m\}$: For failed systems ($\delta_i = 1$), the set of components that could have caused the failure. For censored systems, C_i is undefined.

This data structure is termed *masked data* because the true component cause of failure is not directly observed but is known to belong to the candidate set C_i. The masking mechanism assumes that:

1. The candidate set always contains the true failed component.
2. Given the system failure time and the true failed component, the masking is non-informative (i.e., all candidate sets containing that component are equally likely).

2.4 Well-Designed Systems

A *well-designed series system* is characterized by components having similar but not necessarily identical failure characteristics. Operationally, we define a well-designed system as one where:

1. Component MTTFs are of similar magnitude (within a factor of 2-3).
2. Component shape parameters are reasonably aligned (within approximately 20-30% of each other).
3. No single component dominates as a weak point (i.e., component failure probabilities are relatively balanced).

This concept is important for assessing the appropriateness of reduced models that assume parameter homogeneity.

2.5 Series System Theorems

The following theorems characterize the lifetime distribution of series systems with independent component lifetimes.

Theorem 2.1 (Series System Lifetime)**.** *For a series system of m independent components, the system lifetime T is given by*

$$T = \min\{T_1, T_2, \dots, T_m\},$$

where T_j is the lifetime of component j.

Theorem 2.2 (Series System Reliability Function)**.** *The reliability function of a series system with m independent components is*

$$R(t; \boldsymbol{\theta}) = \prod_{j=1}^{m} R_j(t; \lambda_j, k_j),$$

where $R_j(t; \lambda_j, k_j)$ is the reliability function of component j.

Theorem 2.3 (Series System Hazard Function)**.** *The hazard function of a series system with m independent components is*

$$h(t; \boldsymbol{\theta}) = \sum_{j=1}^{m} h_j(t; \lambda_j, k_j),$$

where $h_j(t; \lambda_j, k_j)$ is the hazard function of component j.

For series systems with Weibull components, these theorems yield specific forms presented in the following subsection.

2.6 Series Systems with Weibull Components

Applying Theorems 2.2, 2.3, and 2.1 to series systems with Weibull-distributed components yields specific analytical forms for the system lifetime distribution.

The lifetime of the series system composed of m Weibull components has a reliability function given by

$$R(t; \boldsymbol{\theta}) = \exp\left\{-\sum_{j=1}^{m} \left(\frac{t}{\lambda_j}\right)^{k_j}\right\}. \tag{6}$$

Proof. By Theorem 2.2,

$$R(t; \boldsymbol{\theta}) = \prod_{j=1}^{m} R_j(t; \lambda_j, k_j).$$

Plugging in the Weibull component reliability functions obtains the result

$$\begin{aligned} R(t; \boldsymbol{\theta}) &= \prod_{j=1}^{m} \exp\left\{-\left(\frac{t}{\lambda_j}\right)^{k_j}\right\} \\ &= \exp\left\{-\sum_{j=1}^{m} \left(\frac{t}{\lambda_j}\right)^{k_j}\right\}. \end{aligned}$$

□

The Weibull series system's hazard function is given by

$$h(t; \boldsymbol{\theta}) = \sum_{j=1}^{m} \frac{k_j}{\lambda_j} \left(\frac{t}{\lambda_j} \right)^{k_j - 1}, \tag{7}$$

whose proof follows from Theorem 2.3.

The pdf of the series system is given by

$$f(t; \boldsymbol{\theta}) = \left\{ \sum_{j=1}^{m} \frac{k_j}{\lambda_j} \left(\frac{t}{\lambda_j} \right)^{k_j - 1} \right\} \exp\left\{ -\sum_{j=1}^{m} (\frac{t}{\lambda_j})^{k_j} \right\}. \tag{8}$$

When components have heterogeneous shape parameters, the series system hazard function can exhibit complex behavior, including both infant mortality (initial decrease) and aging (eventual increase) phases. This bathtub-shaped hazard is commonly observed in engineered systems where early failures due to defects give way to a period of stable operation, eventually followed by wear-out failures.

2.7 Baseline Series System Configuration

Throughout this paper, we use a baseline 5-component series system configuration for simulation studies. The component parameters are specified in Table 1.

Table 1: Baseline 5-Component Series System Parameters

Component j	Shape k_j	Scale λ_j	MTTF_j
1	3.0	30	≈ 26.7
2	2.0	50	≈ 44.3
3	0.5	15	≈ 30.0
4	5.0	20	≈ 18.7
5	0.25	50	≈ 180.0

This baseline system represents a configuration where components 1-4 have relatively similar MTTFs (ranging from approximately 18.7 to 44.3 time units), while component 5 has a substantially longer MTTF. The shape parameters vary from 0.25 to 5.0, spanning decreasing hazard (components 3 and 5), increasing hazard (components 1, 2, and 4), and varying degrees of aging behavior. When component 3 has a shape parameter near 1.1308, the system approximates a well-designed configuration where shape parameters are reasonably aligned.

2.8 Component Failure Probabilities

In series systems, a critical quantity for understanding system behavior and estimator performance is the probability that a particular component causes system failure. Let K_i denote the index of the component that causes the i-th system to fail. The probability that component j is the cause of failure is given by:

$$P_j = \Pr\{K_i = j\} = \int_0^{\infty} f_{T_i, K_i}(t, j; \boldsymbol{\theta})\, dt, \tag{9}$$

where $f_{T_i,K_i}(t, j; \boldsymbol{\theta})$ is the joint density of system lifetime T_i and component cause K_i. This can be expressed as:

$$P_j = \int_0^\infty f_j(t;\theta_j) R_{\setminus j}(t;\boldsymbol{\theta}_{\setminus j})\, dt = E_{\boldsymbol{\theta}} \left\{ \frac{h_j(T_i;\theta_j)}{h(T_i;\boldsymbol{\theta})} \right\}, \tag{10}$$

where $f_j(t;\theta_j)$ is the PDF of component j, $R_{\setminus j}(t;\boldsymbol{\theta}_{\setminus j}) = \prod_{k \neq j} R_k(t;\theta_k)$ is the reliability of all components except j, and $h(t;\boldsymbol{\theta})$ is the system hazard function.

For Weibull components in series, the component failure probability depends on both shape and scale parameters in a complex, non-linear manner. A key insight is that MTTF alone is insufficient for determining failure probabilities in series systems with heterogeneous shape parameters.

Relationship Between Shape Parameter and Failure Probability

When components have different shape parameters, counter-intuitive relationships can arise. Consider a component with shape parameter $k_j < 1$ (decreasing hazard, infant mortality). Such a component may have a *higher* MTTF than components with $k > 1$, yet simultaneously have a *higher* probability of causing system failure. This occurs because:

- Components with $k < 1$ have high early failure rates (infant mortality), making them likely to fail first despite long-term survivors having extended lifetimes.
- Components with $k > 1$ have low early failure rates but increasing hazards (aging), making them less likely to fail first despite lower MTTFs.
- The first failure determines series system lifetime, so early hazard behavior dominates MTTF considerations.

This phenomenon has important implications for MLE behavior and bias patterns. The estimator must balance fitting the observed failure times with correctly attributing failures to the appropriate components. When a component with $k_j < 1$ dominates early failures, the MLE may exhibit bias in shape parameters of other components to compensate for limited information about their failure characteristics.

Implications for Estimation

The component failure probabilities P_j directly influence the information available for estimating each component's parameters:

1. **High failure probability:** Components with higher P_j are observed as the cause of failure more frequently, providing more information for parameter estimation. This typically results in lower estimator variance and better coverage probabilities for that component's parameters.
2. **Low failure probability:** Components with lower P_j are rarely observed as the cause of failure. Parameter estimates for these components have higher variance, wider confidence intervals, and potentially worse coverage properties.
3. **Masking effects:** Candidate sets that include multiple components dilute the information about which component actually failed. The impact is more severe for components with already low P_j.

Throughout the simulation studies in Section 4, we will examine how varying shape and scale parameters affects component failure probabilities and, consequently, estimator performance. Understanding these relationships is essential for interpreting the sensitivity analyses and model selection results that follow.

3 Likelihood Model

This section summarizes the likelihood model for component reliability estimation from masked and right-censored system failure data, as developed in [1]. The key challenge is that only system-level failure times are observed, not individual component failures, and the component cause of failure may be partially masked.

3.1 Model Framework

For the i-th system ($i = 1, \ldots, n$), the observed data consists of:

- System failure or censoring time t_i
- Censoring indicator $\delta_i \in \{0, 1\}$
- Candidate set $C_i \subseteq \{1, 2, \ldots, m\}$ (for failed systems only)

Let $K_i \in \{1, \ldots, m\}$ denote the (unobserved) component cause of failure for system i. The likelihood contribution for a single observation depends on whether the system failed or was censored.

3.2 Likelihood Function Structure

For a censored observation ($\delta_i = 0$), the likelihood contribution is simply the system reliability function evaluated at the censoring time:

$$L_i(\boldsymbol{\theta}|t_i, \delta_i = 0) = R(t_i; \boldsymbol{\theta}) = \prod_{j=1}^{m} \exp\left\{-\left(\frac{t_i}{\lambda_j}\right)^{k_j}\right\}. \tag{11}$$

For a failed system ($\delta_i = 1$) with candidate set C_i, the likelihood contribution accounts for the fact that the failed component is known to be in C_i but is otherwise unknown:

$$L_i(\boldsymbol{\theta}|t_i, \delta_i = 1, C_i) = \sum_{j \in C_i} \Pr\{K_i = j | t_i, \boldsymbol{\theta}\} \cdot f(t_i; \boldsymbol{\theta}), \tag{12}$$

where the conditional probability that component j failed given system failure time t_i is

$$\Pr\{K_i = j | t_i, \boldsymbol{\theta}\} = \frac{h_j(t_i; \lambda_j, k_j)}{h(t_i; \boldsymbol{\theta})} = \frac{h_j(t_i; \lambda_j, k_j)}{\sum_{\ell=1}^{m} h_\ell(t_i; \lambda_\ell, k_\ell)}. \tag{13}$$

The complete log-likelihood for the sample of n systems is

$$\ell(\boldsymbol{\theta}|D) = \sum_{i=1}^{n} \left[(1 - \delta_i) \log R(t_i; \boldsymbol{\theta}) + \delta_i \log\left(\sum_{j \in C_i} \Pr\{K_i = j | t_i, \boldsymbol{\theta}\} \cdot f(t_i; \boldsymbol{\theta})\right)\right]. \tag{14}$$

3.3 Key Assumptions

The likelihood model relies on the following assumptions:

1. **Independence**: System lifetimes are independent and identically distributed.
2. **Series structure**: The system fails when the first component fails.

3. **Weibull components**: Each component lifetime follows a two-parameter Weibull distribution.

4. **Candidate set validity**: For failed systems, the candidate set C_i always contains the true failed component, i.e., $K_i \in C_i$.

5. **Non-informative masking**: Given the system failure time t_i and the true failed component K_i, the masking mechanism that determines C_i is non-informative about the parameters $\boldsymbol{\theta}$.

3.4 Maximum Likelihood Estimation

Maximum likelihood estimates are obtained by maximizing the log-likelihood function $\ell(\boldsymbol{\theta}|D)$ with respect to $\boldsymbol{\theta}$. Due to the complexity of the likelihood surface with $2m$ parameters, numerical optimization is required. The optimization is performed using the L-BFGS-B algorithm [2], which handles box constraints on the parameters (all shape and scale parameters must be positive).

Previous simulation studies [1] demonstrated that maximum likelihood estimation produces accurate results despite small samples and significant masking and censoring. However, shape parameters exhibit greater variability and are more challenging to estimate precisely than scale parameters, particularly when candidate sets are large or when certain components rarely fail.

4 Simulation Study: Sensitivity Analysis to Changing System Design

This section presents simulation studies assessing the sensitivity of maximum likelihood estimators to deviations from the baseline system configuration. We examine how changes in individual component parameters affect estimator performance, bias, dispersion, and coverage probability.

4.1 Simulation Methodology

For each simulation scenario, we employ the following methodology:

- **Number of replications**: 1000 Monte Carlo replications per parameter configuration
- **Sample size**: $n = 100$ systems per replication (unless otherwise specified)
- **Masking probability**: $p = 0.215$ (moderate masking, unless otherwise specified)
- **Censoring mechanism**: Right-censoring at the $q = 0.825$ quantile of the system lifetime distribution (moderate censoring, unless otherwise specified)
- **Candidate set generation**: For each failed system, components are independently included in the candidate set with probability p, ensuring the true failed component is always included
- **Optimization**: L-BFGS-B algorithm [2] with box constraints requiring all parameters to be positive
- **Confidence intervals**: Bias-corrected and accelerated (BCa) bootstrap confidence intervals [3] with 2000 bootstrap samples per replication, targeting 95% nominal coverage
- **Performance metrics**: Bias, dispersion (interquartile range of point estimates), coverage probability, and confidence interval width

The term *moderate* refers to levels that are substantial enough to pose estimation challenges but not so extreme as to make inference infeasible. Specifically, a masking probability of 0.215 results in candidate sets containing approximately 2-3 components on average, and a censoring quantile of 0.825 censors approximately 17.5% of observations.

4.2 Scenario: Assessing the Impact of Changing the Scale Parameter of Component 3

By Equation 5, we see that MTTF_j is proportional to the scale parameter λ_j, which means when we decrease the scale parameter of a component, we proportionally decrease the MTTF. In this scenario, we start with the well-designed series system described in Table 1, and we will manipulate the MTTF of component 3, MTTF_3, by changing its scale parameter, λ_3, and observing the effect this has on the MLE. Since the other components had a similiar MTTF, we will arbitrarily choose component 1 to represent the other components. The bottom plot shows the coverage probabilities for all parameters.

In Figure 1, we show the effect of changing the scale parameter of component 3, $lambda_3$, but map λ_3 to MTTF_3 to make it more intuitive to reason about. We vary the MTTF of component 3 from 300 to 1500 and the other components have their MTTFs fixed at around 900, as shown in Table 1. We fix the masking probability to $p = 0.215$ (moderate masking), the right-censoring quantile to $q = 0.825$ (moderate censoring), and the sample size to $n = 100$ (moderate sample size).

Key Observations

Coverage Probability (CP) When MTTF of component 3 is much smaller than other components, the CP for k_3 is very well calibrated (approximately obtaining the nominal level 95%) while the CP for other componentns are around 90%, which is still reasonable. (This is the case even though the width of the CI for k_3 is extremely narrow compared to the others). As MTTF_3 increases, the CP for k_3 decreases, while the CP for the other components increase slightly. The scale parameters are generally well-calibrated for all of the components, except for component 3 when its MTTF is large and it dips down to 90%. Despite the individual differences, the mean of the CPs for shape and scale parameters hardly change.

Dispersion of MLEs For component 3, as its MTTF decreases, the dispersion of MLEs narrows, indicating more precise estimates. Conversely, dispersion for other components widens. As MTTF of component 3 increases, its dispersion widens while others narrow. This is consistent with the fact that the smaller MTTF of component 3 means that, in this well-designed system at least, it is more likely to be the component cause of failure, and so we have more information about its parameters and are able to estimate them more accurately.

IQR of Bootstrapped CIs The dark blue vertical lines representing IQR are consistent with the dispersion of MLEs, which is the ideal behavior, and suggests that the BCa confidence intervals are performing well.

Bias of MLEs For component 3, the bias of MLE for the scale parameter becomes slightly more negatively biased as MTTF_3 increases, and the bias of the MLE for the shape parameter becomes slightly more positively biased. The MLE for the shape and scale parameters for component 1 have a very small bias, if any, and are not affected by the MTTF_3. The scale parameters are easier to

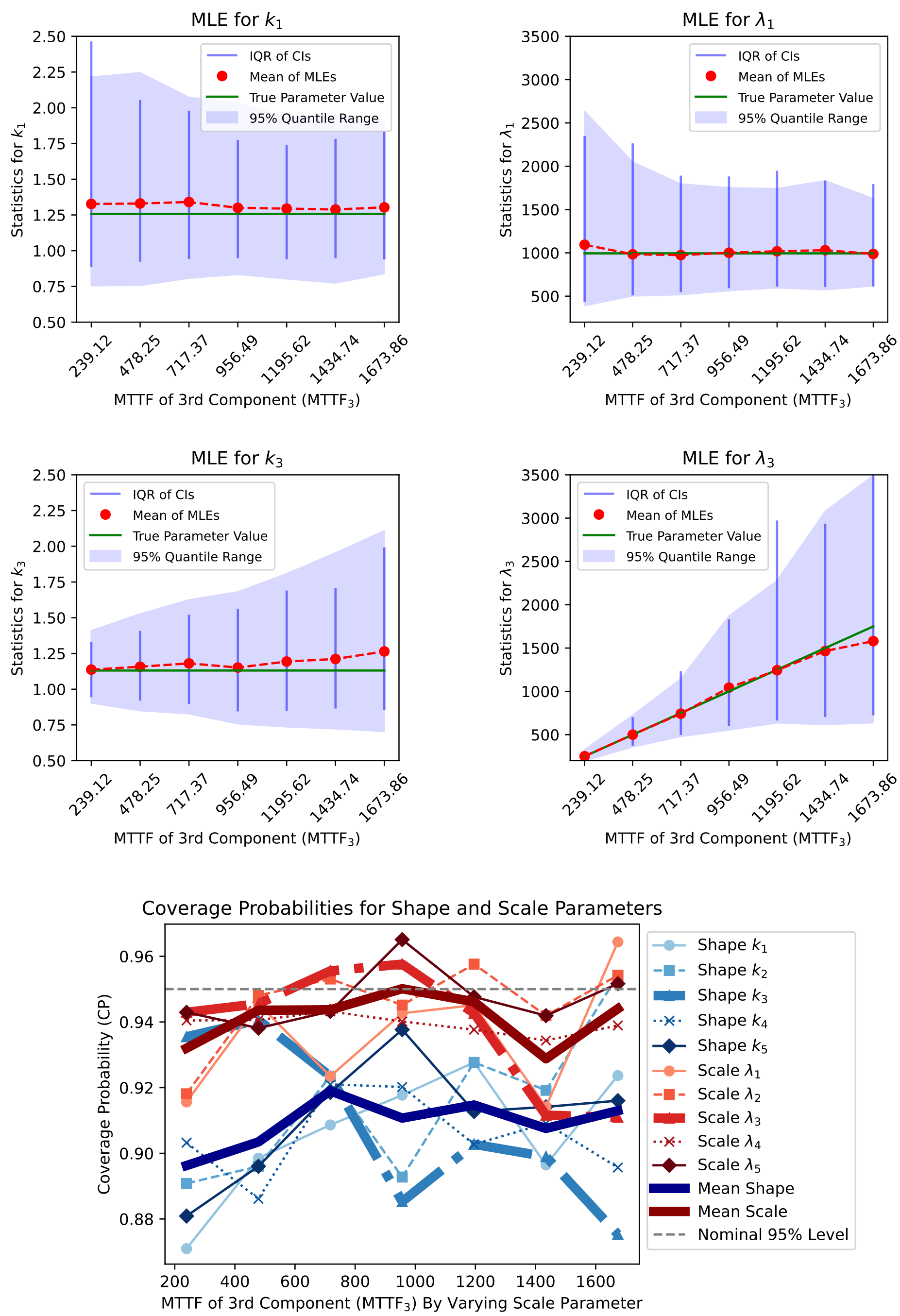

Figure 1: MTTF of Component 3 vs MLE By Varying Scale

estimate than the shape parameters, and so they are less sensitive to changes in scale than the shape parameters, as we will show in the next scenario.

4.3 Scenario: Assessing the Impact of Changing the Shape Parameter of Component 3

The shape parameter determines the failure characteristics. We vary the shape paramenter of component 3 from 0.1 to 3.5 and observe the effect it has on the MLE. When $k_3 < 1$, this indicates infant mortality, and when $k_3 > 1$, this indicates wear-out failures.

We analyze the effect of component 3's shape parameter on the MLE and the bootstrapped confidence intervals for the shape and scale parameters of components 1 and 3 (the component we are varying). First, we look at the effect on the scale parameter.

Key Observations

Coverage Probability (CP) The CP for the scale parameters are well-calibrated and close to the nominal level of 0.95 for all values of $\Pr\{K_i = 3\}$. For the the shape parameter of component 3 (k_3) in bold orange colors, we see that it is well-calibrated for all values of $\Pr\{K_i = 3\}$, but actually may become too large for extreme values of $\Pr\{K_i = 3\}$. The CP for the shape parameters of the other components decreases with $\Pr\{K_i = 3\}$, dipping below 90% for $\Pr\{K_i = 3\} > 0.4$. At a sample size of $n = 100$, the CP for the shape parameters of the other components is generally not well-calibrated for $\Pr\{K_i = 3\} > 0.4$.

Dispersion of MLEs The dispersion of the MLE for the shape and scale parameters of component 1, k_1 and λ_1, is fairly steady but begins to increase rapdily at the extreme values of $\Pr\{K_i = 3\}$. This is indicative of having less information about the failure characteristcs of component 1 as component 3 begins to dominate the component cause of failure. The dispersion of the shape parameter k_3 is initially quite large, indicative of having very little information about the failure characteristcs of component 3 since it is unlikely to be the component cause of failure, but its dispersion rapidly decreases as $\Pr\{K_i = 3\}$ increases and more information is available about component 3's failure characteristics. In fact, it nearly becomes a point at $\Pr\{K_i = 3\} = 0.6$. The dispersion of the the scale parameter of component 3, λ_3, is quite steady and is less spread out than the MLE for λ_1, but at extreme values of $\Pr\{K_i = 3\}$, it also begins to rapidly increase, suggesting some complex interactions between the shape and scale parameters of component 3.

IQR of Bootstrapped CIs The CIs precisely track the dispersion of the MLEs, which is the ideal behavior, and suggests that the BCa confidence intervals are performing well.

Bias of MLEs The MLE for the scale parameters are nearly unbiased and generally seem unaffected by changes in $\Pr\{K_i = 3\}$. As $\Pr\{K_i = 3\}$ increases the MLE is adjusting k_1 to be more positively biased, decreasing its infant morality rate to make it less likely to be the component cause of failure, and adjusting k_3 to be less positively biased, increasing its infant mortality rate, to make it more likely to be the component cause of failure.

5 Weibull Series Homogenous Shape Model

In the sensitivity analysis in Section 4, we found that the MLE is very sensitive to deviations from a well-designed system. In this section, we develop a reduced model that assumes homogeneity in the

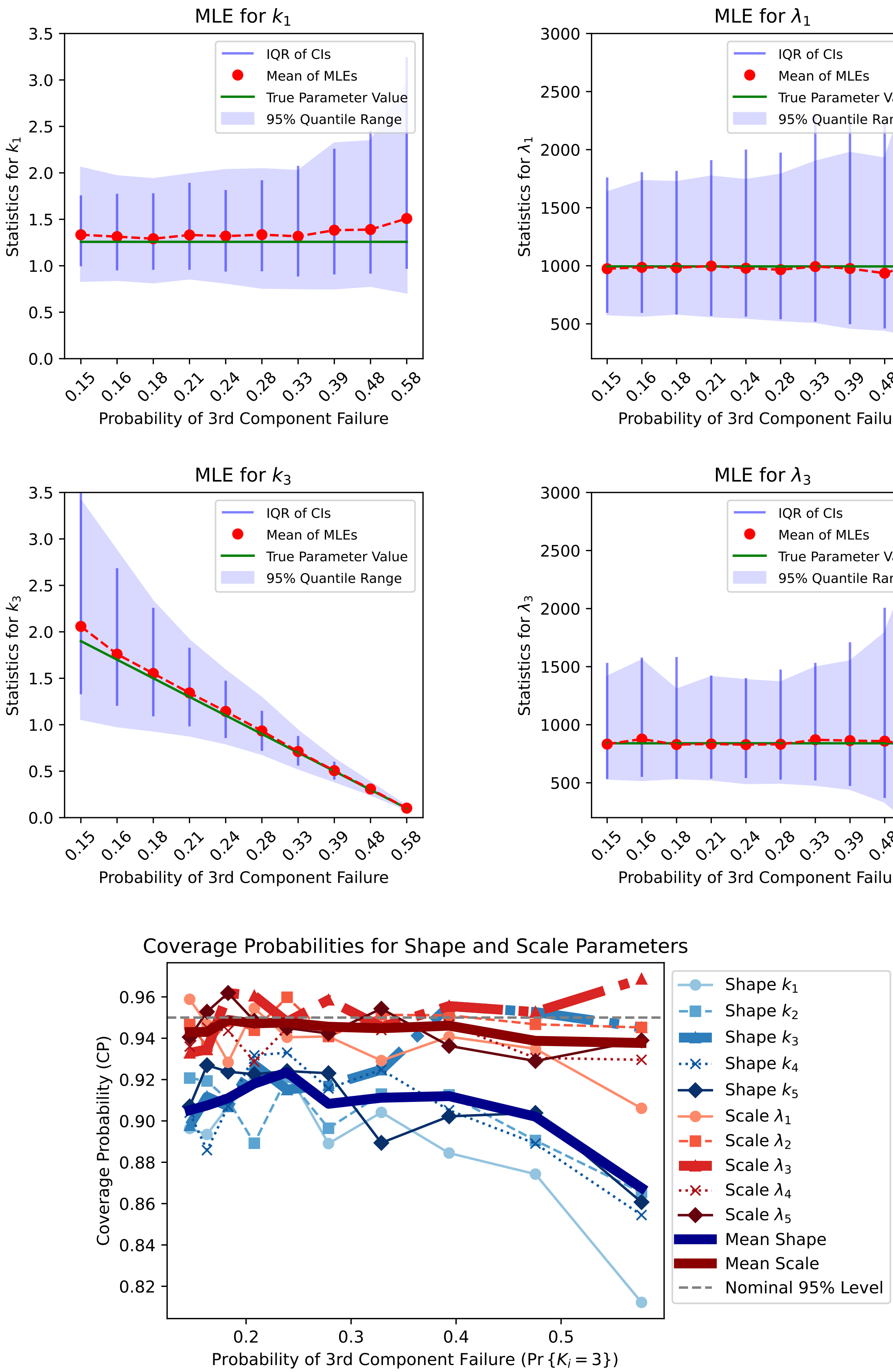

Figure 2: Probability of Component 3 Failure vs MLE

shape parameters of the components, which simplifies analysis and reduces estimator variability.

Here, our focus shifts to a sensitivity analysis aimed at understanding when it is appropriate to use the reduced model that assumes homogeneity in the shape parameters of the components. The reduced model offers interpretability (the series system is itself Weibull) and reduced estimator variability (only $m+1$ parameters instead of $2m$), but it must adequately describe the data.

5.1 Homogeneous Shape Model Definition

The *homogeneous shape model* (also called the *reduced model*) assumes that all components share a common shape parameter k while retaining individual scale parameters λ_j. The parameter vector for the reduced model is

$$\boldsymbol{\theta}_R = (k, \lambda_1, \lambda_2, \ldots, \lambda_m), \tag{15}$$

reducing the number of parameters from $2m$ in the full model to $m+1$ in the reduced model.

Under this assumption, each component has a Weibull lifetime with

$$T_{ij} \sim \text{Weibull}(k, \lambda_j), \quad j = 1, \ldots, m. \tag{16}$$

A key property of the homogeneous shape model is that the series system lifetime distribution is itself Weibull. Specifically, for the system lifetime $T_i = \min\{T_{i1}, \ldots, T_{im}\}$, we have

$$T_i \sim \text{Weibull}\left(k, \lambda_s\right), \tag{17}$$

where the system scale parameter is given by

$$\lambda_s = \left(\sum_{j=1}^{m} \lambda_j^{-k}\right)^{-1/k}. \tag{18}$$

This Weibull property of the system lifetime provides several advantages:

1. **Analytical tractability**: System reliability metrics (MTTF, reliability function, hazard function) have closed-form expressions.

2. **Interpretability**: The system exhibits a single failure mode characterized by the common shape parameter k.

3. **Reduced variance**: Fewer parameters to estimate results in lower estimator variance, particularly for the shape parameter.

4. **Computational efficiency**: Optimization is faster with $m+1$ parameters instead of $2m$ parameters.

However, the reduced model is appropriate only when components genuinely have similar failure characteristics. When component shape parameters differ substantially, forcing homogeneity can lead to poor model fit and biased parameter estimates.

5.2 Assessing the Appropriateness of the Reduced Model

In order to determine if a reduced model (e.g., Weibull series system in which all of the shape parameters are homogeneous) is appropriate, a hypothesis test test may be conducted to determine if there is statistically significant evidence in support of the null hypothesis H_0, e.g., that all of the shape parameters are homogeneous.

The likelihood function of the reduced model is related to the likelihood function of the full model. We denote the full model likelihood function by L_F and the reduced model likelihood by L_R. The reduced model is obtained by setting the shape parameter of each component to be the same, i.e., $k_1 = \cdots = k_m = k$. Thus, the reduced model likelihood function is given by

$$L_R(k, \lambda_1, \lambda_2, \cdots, \lambda_m | D) = L_F(k, \lambda_1, k, \lambda_2, \ldots, k, \lambda_m | D),$$

The same may be done for the score and hessian of the log-likelihood functions.

Given that we employ a well-defined likelihood model, the likelihood ratio test (LRT) is a good choice. The LRT statistic is given by

$$\Lambda = -2(\log L_R(\hat{\theta}_R | D) - \log L_F(\hat{\theta} | D))$$

where L_R is the likelihood of the reduced (null) model evaluated at its MLE $\hat{\theta}_R$ given a random sample D of masked data and L_F is the likelihood of the full model evaluated at its MLE $\hat{\theta}$ given the same set of data D. Under the null model, the LRT statistic is asymptotically distributed chi-squared with $m - 1$ degrees of freedom, where m is the number of components in the series system,

$$\Lambda \sim \chi^2_{m-1}.$$

If the LRT statistic is greater than the critical value of the chi-squared distribution with $m - 1$ degrees of freedom, $\chi^2_{m-1,1-\alpha}$, where α denotes the significance level, then we find the data to be incompatible with the null hypothesis H_0.

5.3 Simulation Study: Full Weibull Model vs Reduced (Homogenous Shape) Model

We aim to assess the appropriateness of the reduced model under varying sample sizes and shape parameters of the third component (k_3). We employ a simulation study using the likelihood ratio test (LRT) for this purpose, where the null hypothesis, H_0, assumes homogenous shape parameters.

We take the well-designed series system described in Table 1, and manipulate the shape parameter of the third component (k_3) to cause the components to have different failure characteristics. Recall that $k_3 = 1.1308$ corresponds to a *well-designed* series system, where component shapes are reasonably aligned. We also vary the sample size n to assess the impact of sample size on the appropriateness of the reduced model.

Figure 3 provides a contour plot with varying sample sizes along the x-axis, shape of component 3 along the y-axis, and median p-value along the color scale. The contour lines corresponding to p-values of 0.05 and 0.1 are often used as a threshold for statistical significance. Points outside of these contours in dark blue are indicative of significant evidence against the null model and points inside of these contours in light blue for 0.1 and green for 0.05 are indicative of insufficient evidence against the null model.

Figure 4 provides a plot of the median p-value against the sample size for the well-designed system, where the shape parameter of component 3 is fixed at 1.1308. The 95th percentile of the p-values is also provided as a more stringent criterion for statistical significance.

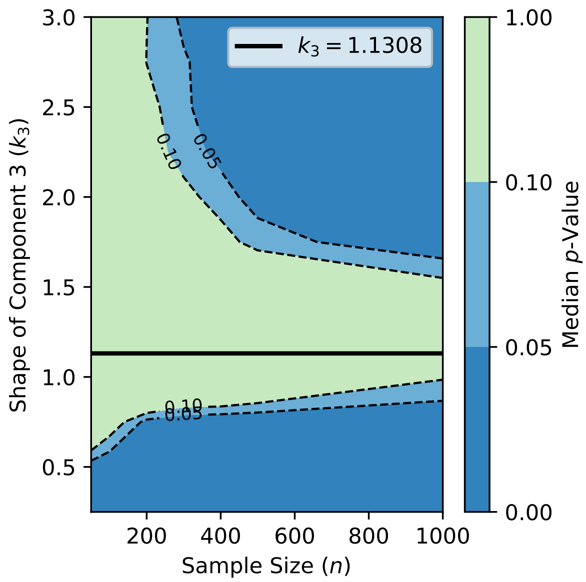

Figure 3: p-Value vs Sample Size and Shape k_3

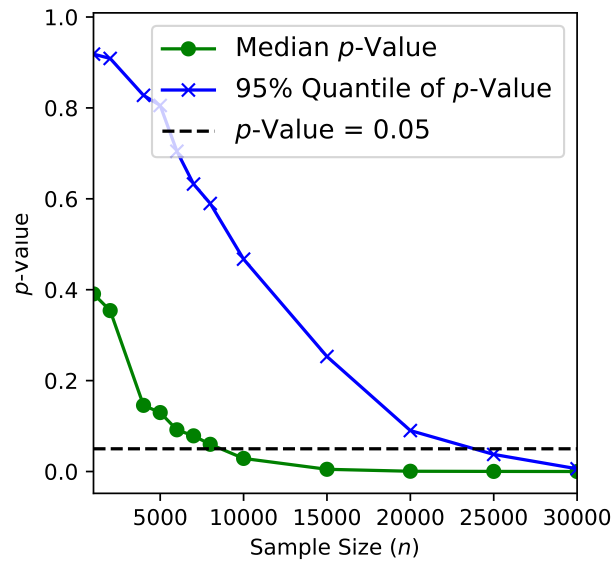

Figure 4: p-Value vs Sample Size for Well-Designed System

Sensitivity to Sample Size (n)

- The sample size is an essential aspect of hypothesis testing, as it affects the test's power, which is the probability of correctly rejecting the null hypothesis when it is false. In the contour plot in Figure 3, as n increases, the contours trend lower. This indicates that larger samples result in smaller median p-values, implying that the power of the LRT increases with the sample size. However, its power is quite low for small samples, particularly for values of k_3 somewhat close to the shape parameters of the other components in the system.

- Recall that in the well-designed series system, $k_3 = 1.1308$. In this case, even very large sample sizes do not produce evidence against the null model, indicating robust compatibility.

- In Figure 4, we fix k_3 at 1.1308 and vary the sample size. The median p-value only manages to drop below the 0.05 theshold with sample sizes around 10000. In the more stringent criterion given by the 95th percentile of the p-values, nearly 30000 observations are necessary to reject the null hypothesis in 95% of the simulations.

Sensivity to Shape Parameter (k_3)

- In Figure 3, for a given shape parameter, increasing the sample size tends to decrease the median p-value. Larger samples provide more information about the parameters, which increases the power of the LRT.

- The median p-values in the vicinity of the line $k_3 = 1.1308$ are high across various sample sizes, indicating that the null model is a good fit. As k_3 deviates from this line, the median p-value diminishes, indicating increasing evidence against the null model.

5.4 Implications and Recommendations

The power of the test for a well-designed series system is quite low, requiring many thousands of observations before the test has sufficient power to reject the null hypothesis. But, this is not necessarily a bad thing. The reduced model is quite simple and interpretable, and is by definition a good fit for a well-designed series system.

The findings suggest that the reduced model is particularly apt when the system is well-designed, even for very large samples. Practitioners should weigh the trade-offs between the simplicity of the reduced model and the adequacy in describing the data, with consideration of the available sample size and the characteristics of the system being modeled.

For systems believed to be well-designed, employing the null model is supported both statistically and practically due to its simplicity, reduced estimator variability, and analytical tractability. In the absence of prior information, or if the shape parameter significantly diverges from the well-designed value, the choice between models should be undertaken with caution. More complex models may be favorable, especially with large sample sizes.

6 Conclusion

In this study, we employed simulation techniques and Likelihood Ratio Tests (LRTs) to assess the adequacy and sensitivity of Maximum Likelihood Estimators (MLEs) for reliability assessment in 5-component series systems with Weibull component lifetimes. Two main models were examined: a more complex model with heterogeneous shape parameters and a reduced model assuming homogeneous shape parameters for all components.

The reduced model improves interpretability by rendering the system Weibull and reduces estimator variability, thus appearing statistically and practically favorable for well-designed systems. These well-designed systems are characterized by similar but non-identical failure characteristics among components, without a single weak point. Even for large samples, the reduced model showed excellent fit in these cases.

However, our simulations revealed that varying a single component's scale or shape parameter quickly provided evidence against the reduced model's adequacy. This suggests that more complex models may be preferable in systems with divergent component properties or when sample sizes are large.

Estimator performance was found to be sensitive but robust, particularly concerning the challenges introduced by limited, right-censored, and masked failure data. Bootstrap confidence intervals proved valuable in characterizing estimator uncertainty. As the sample size increased, estimator dispersion reduced, and confidence intervals narrowed, although small samples exhibited bias, particularly in shape parameters. Masking probability also played a role, expanding confidence intervals to maintain coverage while largely leaving scale parameters unbiased.

In summary, the choice between the reduced and more complex models should weigh the trade-offs between simplicity and representativeness. Our findings offer practical guidance for reliability assessments, particularly when dealing with limited system failure data. Proper model specification ultimately requires a nuanced understanding of both system characteristics and estimator behavior.

A Parameter Sensitivity Analysis Tables

This appendix provides detailed tables quantifying the relationships between Weibull parameters and component failure probabilities for a simplified 3-component series system. These tables illustrate the complex, non-linear relationships between shape parameters, scale parameters, mean time to failure (MTTF), and the probability of each component causing system failure.

A.1 Effect of Varying Shape Parameter

Table 2 shows the effect of varying the shape parameter of component 1 (k_1) from 0.1 to 1.0 while holding $k_2 = k_3 = 0.5$ and all scale parameters at $\lambda_1 = \lambda_2 = \lambda_3 = 1$.

Table 2: Effect of Varying Shape Parameter k_1 on Failure Probabilities and MTTFs

k_1	P_1	P_2	P_3	MTTF_1	MTTF_2	MTTF_3	System MTTF
0.10	0.77	0.12	0.12	10.00	2.00	2.00	0.89
0.20	0.69	0.15	0.15	4.59	2.00	2.00	1.02
0.30	0.64	0.18	0.18	3.32	2.00	2.00	1.10
0.40	0.59	0.20	0.20	2.68	2.00	2.00	1.17
0.50	0.55	0.22	0.22	2.29	2.00	2.00	1.22
0.60	0.52	0.24	0.24	2.03	2.00	2.00	1.27
0.70	0.48	0.26	0.26	1.85	2.00	2.00	1.31
0.80	0.45	0.27	0.27	1.71	2.00	2.00	1.34
0.90	0.42	0.29	0.29	1.61	2.00	2.00	1.38
1.00	0.40	0.30	0.30	1.53	2.00	2.00	1.41

Key observations from Table 2:

- As k_1 increases from 0.1 to 1.0, the failure probability P_1 decreases from 0.77 to 0.40, while P_2 and P_3 increase proportionally.
- Component 1 has the *highest* MTTF when $k_1 = 0.1$ ($\mathrm{MTTF}_1 = 10.0$), yet also has the *highest* failure probability ($P_1 = 0.77$). This counter-intuitive result demonstrates that MTTF alone is insufficient for predicting failure probabilities in series systems with heterogeneous shape parameters.
- The shape parameter dominates early hazard behavior. Components with $k < 1$ exhibit high infant mortality, making them likely to fail first despite having longer MTTFs than components with $k > 1$.
- System MTTF increases as k_1 approaches 1.0, reflecting reduced infant mortality in the overall system.

A.2 Effect of Varying Scale Parameter

Table 3 shows the effect of varying the scale parameter of component 1 (λ_1) from 1 to 4 while holding all shape parameters at $k_1 = k_2 = k_3 = 0.5$ and $\lambda_2 = \lambda_3 = 1$.

Table 3: Effect of Varying Scale Parameter λ_1 on Failure Probabilities and MTTFs

λ_1	P_1	P_2	P_3	MTTF_1	MTTF_2	MTTF_3	System MTTF
1.0	0.55	0.22	0.22	2.00	2.00	2.00	1.22
2.0	0.35	0.32	0.32	4.00	2.00	2.00	1.72
3.0	0.25	0.38	0.38	6.00	2.00	2.00	1.98
4.0	0.20	0.40	0.40	8.00	2.00	2.00	2.14

Key observations from Table 3:

- Unlike the shape parameter, the scale parameter exhibits a more intuitive relationship: as λ_1 increases, MTTF_1 increases proportionally and P_1 decreases.
- When shape parameters are homogeneous ($k_1 = k_2 = k_3 = 0.5$), the component with the largest scale parameter has the lowest failure probability, and MTTF is directly proportional to the scale parameter.
- System MTTF increases with λ_1, but at a decreasing rate due to the series configuration (weakest link).
- This more linear relationship makes scale parameters easier to estimate than shape parameters, which is consistent with observations in the simulation studies.

A.3 Joint Variation of Shape and Scale Parameters

Table 4 shows the joint effect of varying both k_1 and λ_1 simultaneously, demonstrating the complex interactions between these parameters.

Table 4: Joint Effect of Varying Both k_1 and λ_1 on Failure Probabilities

k_1	λ_1	P_1	P_2	P_3	MTTF_1	MTTF_2	MTTF_3	System MTTF
0.25	1.0	0.64	0.18	0.18	3.63	2.00	2.00	1.10
0.25	2.0	0.50	0.25	0.25	7.26	2.00	2.00	1.43
0.25	3.0	0.41	0.29	0.29	10.89	2.00	2.00	1.62
0.50	1.0	0.55	0.22	0.22	2.00	2.00	2.00	1.22
0.50	2.0	0.35	0.32	0.32	4.00	2.00	2.00	1.72
0.50	3.0	0.25	0.38	0.38	6.00	2.00	2.00	1.98
0.75	1.0	0.48	0.26	0.26	1.40	2.00	2.00	1.31
0.75	2.0	0.25	0.38	0.38	2.80	2.00	2.00	1.98
0.75	3.0	0.16	0.42	0.42	4.19	2.00	2.00	2.32

Key observations from Table 4:

- For a fixed shape parameter k_1, increasing λ_1 decreases P_1 (component 1 becomes less likely to fail first).
- For a fixed scale parameter λ_1, increasing k_1 also decreases P_1, but the mechanism is different: higher k_1 reduces infant mortality.
- The joint effects are multiplicative rather than additive. A component with low shape ($k_1 = 0.25$) and high scale ($\lambda_1 = 3.0$) still has MTTF $= 10.89$ and $P_1 = 0.41$, demonstrating the dominance of early hazard behavior in series systems.
- To minimize a component's failure probability, both increasing its scale parameter and increasing its shape parameter toward 1.0 are effective strategies, but they work through different mechanisms.

A.4 Implications for Model Selection and Estimation

These tables quantitatively demonstrate several key principles that inform the simulation studies and model selection analyses in the main text:

1. **MTTF is insufficient:** Component failure probabilities depend on the entire hazard function shape, not just the mean. Systems with heterogeneous shape parameters require careful analysis beyond MTTF comparisons.

2. **Shape parameter complexity:** The non-linear relationship between shape parameters and failure probabilities explains why shape parameters are harder to estimate and exhibit greater bias than scale parameters.

3. **Information availability:** Components with high failure probabilities provide more data for estimation. When P_j is small, parameter estimates for component j will have high variance regardless of sample size.

4. **Model selection criteria:** The reduced model (homogeneous shapes) is appropriate when shape parameters are already similar, as failure probabilities become more predictable from MTTF alone. When shape parameters diverge substantially, the full model is necessary to capture the complex failure probability structure.

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